

EBN111 SEMESTER TOETS/TEST 2 (8 MEI/MAY 2013)
ANTWOORDBLAD / ANSWER SHEET

Van / Surname:	MEMO	Studentenommer: Student number:							
Voorletters / Initials:		Tel. nr. / Tel no.:							
Dissipline / Discipline:		Taal / Language:				Afr	Eng		

Q1		Q2		Tot:	/50
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VRAAG 1 – QUESTION 1 (25)		
1.1	$V_S = 17.5 \text{ V}$	$I_S = 0.125 \text{ A}$
1.2	$a = -40$	$b = 180$
	$c = -80$	$d = 20$
1.3	$\begin{bmatrix} 0.003 & -0.001 \\ -0.001 & 0.005 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -0.009 \\ 0.01 \end{bmatrix}$	
1.4	$V_0 / V_S = 0.125$	
1.5	$a) V_S = 97 \text{ V}$	$b) V_{CE} = -4.8 \text{ V}$
1.6	$a) I_{O(120V)} = -1 \text{ mA}$	$b) I_{O(1mA)} = 0.5 \text{ mA}$
1.7	$I_X = 0.4 \text{ A}$	
1.8	$a) R = 20 \Omega$	$b) I_S = f(i_x) = 2 + 3i_x$
1.9	$a) V_S = 10 \text{ V}$	$b) V_0 = 15 \text{ V}$

VRAAG 2 – QUESTION 2 (25)

2.1	$V_{TH} = 3\text{ V}$	$R_{TH} = 3.33\ \Omega$	
2.2	$I_N = 3\text{ A}$	$R_N = 0.4\ \Omega$	
2.3	$R_2 = 10\ \Omega$	$V_{Th} = 28\text{ V}$	$R_l = 14\ \Omega$
2.4	$v^- = 5\text{ V}$	$v_0 = 20\text{ V}$	$i_l = 2.5\text{ mA}$
2.5	$R_f = 50\text{ k}\Omega$	$v^- = R_b / (R_a + R_b) * V_b$	
	$R_a = 20\text{ k}\Omega$	$R_b = 200\text{ k}\Omega$	
2.6	$a) v_0 = -16\text{ V}$	$b) v_0 = -3\text{ V}$	$c) v_0 = 10\text{ V}$
2.7	$i_l = 42.9\text{ }\mu\text{A}$	$v_2 = -11.5\text{ V}$	
	$v_3/v_2 = 1$	$v_4/v_3 = -2.14$	